

Class of 2026 · Applying directly for a fully-funded PhD in Sustainable Energy (Australia)



徐其胜

QISHENG XU

B.Eng. (2026) in Safety Engineering · Fire Dynamics & Energy Systems Safety

My research uses fire dynamics simulation and quantitative risk assessment to study the initiation, propagation and control boundaries of thermal runaway in energy storage — from device-level thermal stability to enclosure-scale safety-spacing criteria — producing results that can directly inform engineering design and standards.

FDS fire modelling Thermakin pyrolysis QRA Battery thermal runaway

Safety spacing criteria

Fuzhou University · Safety Engineering

Available from Jul 2026

First Prize, National Undergraduate Competition (Aug 2026)

Outstanding Undergraduate Thesis

Monash Master's offer held

PTE Academic 60 (equivalent to IELTS 6.5)

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1	1	5	2
Journal Articles	Patents	Software Copyrights	National Awards
Fuzhou University Undergraduate Thesis (Outstanding, university level) ·	1 first author	2 first author	1 1st + 1 2nd

EDUCATION

Education

Monash University **Master of Applied Engineering (admitted, not enrolled)**

2026 intake

Admitted

Sustainable Energy · Melbourne, Australia

Offer received and deposit paid; place currently held. Applying directly for doctoral study and therefore not yet enrolled.

Fuzhou University

B.Eng.

Sep 2022 — Jun 2026

Safety Engineering · Fuzhou, Fujian, China

🎓 Core-course GPA 4.3/5.0, ranked 1/52

📄 Undergraduate Thesis

Pyrolysis and Combustion Experiments and Numerical Simulation of Flame-Retardant Bamboo
(阻燃竹材的热解和燃烧实验及数值模拟研究)

Supervisor: Prof. Fuqiang Yang Fuzhou University Outstanding Undergraduate Thesis

Core Coursework

Advanced Mathematics		University Physics	
Fluid Mechanics		Engineering Thermodynamics	
Heat Transfer		Combustion Science	
Occupational Health Engineering	96	Safety Education and Training	95
Safety Systems Engineering	92	Safety Assessment	92
Building Safety	92		

Academic Honors

- National Inspirational Scholarship (multiple times)
- University Special Scholarship, Top 1% (multiple times)
- Fuzhou University "Triple-A" Student

RESEARCH INTERESTS

Research Interests

01 Sustainable Energy & Storage Safety (primary focus)

System safety across the full chain of battery energy storage: from device-level thermal stability and module-scale thermal runaway propagation to enclosure-scale safety-spacing criteria. Combining numerical simulation with quantitative risk assessment to build methods that can directly inform engineering design and standards; my undergraduate flame-retardant pyrolysis modelling can extend down to battery-material heat-release characterisation. Australia, with its world-leading renewable penetration and utility-scale storage rollout, is an ideal setting to validate such methods.

Energy storage safety

Thermal runaway propagation

Device-level thermal stability

Safety-spacing criteria

Energy transition

02 Fire Dynamics and Numerical Simulation

Fire field modelling with FDS and fire spread mechanisms, with a focus on heat release rate prediction and smoke transport and control; material combustion characterisation and fire prediction using Thermakin.

FDS Thermakin Fire spread

Smoke control

03 **Quantitative Risk Assessment of Energy Systems**

Quantitative risk assessment (QRA) for energy and industrial systems: hazard identification under the dual-prevention mechanism, risk-graded control, closed-loop hazard screening, and safety system resilience.

QRA Dual-prevention mechanism

Risk grading Resilience

04 **Occupant Egress and Emergency Optimisation**

Multi-scenario emergency evacuation simulation with Pathfinder, focusing on crowd organisation and smart egress guidance; related work won first and second prizes at national competitions.

Pathfinder Evacuation simulation

Crowd dynamics

PUBLICATIONS

Publications

Journal Articles 1

[1] completed

Pyrolysis and Combustion Experiments and Numerical Simulation of Flame-Retardant Bamboo

Qisheng Xu (sole author) (Advisor: Prof. Fuqiang Yang; experiments + numerical simulation validating the "device-level characterisation → numerical extrapolation" workflow)

Fuzhou University Undergraduate Thesis (Outstanding, university level)

Fuzhou University Undergraduate Thesis (Outstanding, university level)

Patents & Software Copyrights 6 (3 as first author)

Patent **An Exit-Indication Terminal and Evacuation System (一种出口指示终端及疏散系统)**

First author

Aug 2025

Software

Graded Control Software for Occupational Health Physical Hazards (职业卫生物理危害分级管控软件)

First author

Software

Dynamic Monitoring Service Software for On-Duty Personnel (在岗人员动态监测服务软件)

First author

Software

Dual Verification System for High-Risk Operator Qualification and Identity (高危作业人员资质与身份双验证系统)

Oct 2025

Software

Remote-Sensing Monitoring and Assessment System for the Ecological Environment of Construction Sites (建筑施工场地生态环境遥感监测与评价系统)

Second author

Software

Regional Carbon Emission Estimation and Environmental Safety Assessment System Based on Remote Sensing Imagery (基于遥感影像的区域碳排放估算与环境安全评估系统)

Fourth author

RESEARCH EXPERIENCE

Research Experience

Design of a Dual-Prevention Mechanism for Fuzhou Library Project Leader

2025.04 - 2025.06

Fuzhou Library

Safety management system optimisation for a public cultural venue, establishing a dual-prevention mechanism of hazard identification and risk screening.

- Served as project leader: overall planning, task allocation and cross-department coordination, delivering all outputs on schedule.
- Led the team in designing and implementing the dual-prevention mechanism (hazard identification and risk screening), improving the library safety management system and reducing latent hazards.
- Designed a risk-graded control and hazard-screening scheme, forming a closed-loop "identification – assessment – response" management model.
- Applied SCL and JHA for hazard identification and the LS and LEC methods for risk grading, coordinating with functional departments to implement the scheme.

SCL	JHA	LS method	LEC method
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Fire Safety Evacuation Design for the Ta Lane Historic District, Three Lanes and Seven Alleys, Fuzhou

Project Leader

2024.12 - 2025.01

Three Lanes and Seven Alleys Historic District, Fuzhou

Fire evacuation modelling and scheme design addressing fire risk and egress difficulties in a historic district.

- Served as project leader: oversaw the full workflow of field survey, modelling and scheme design, and coordinated with district authorities to put the results into practice.
- Built the district evacuation model, produced building CAD drawings and 3D models, and simulated multi-scenario emergency evacuation with Pathfinder.
- Following the General Code for Fire Protection of Buildings and other standards, optimised escape routes and safety facilities for different occupant groups through field survey and data analysis.
- Conducted qualitative and quantitative fire risk analysis, designed an evacuation route optimisation scheme, and coordinated with relevant departments to ensure implementation.

Pathfinder

AutoCAD / 3D modelling

Building fire codes

Quantitative risk analysis

HONORS & AWARDS

Honors & Awards

First Prize, 12th National Undergraduate Practice and Innovation Works Competition in Safety Science and Engineering

National

Aug 2026

Team leader and first author

"Multimodal AI-Interactive Detection Robot System" (多模态 AI 交互探测机器人系统)

The highest-level undergraduate discipline competition in safety science and engineering in China. The work integrates multimodal sensing (vision/gas/temperature) into a disaster-site detection robot, applicable to fire/energy-storage incident reconnaissance and early hazard identification — directly relevant to early-stage thermal runaway detection.

Second Prize, 11th National Undergraduate Practice and Innovation Works Competition in Safety Science and Engineering		
National		Aug 2025
Team leader and first author		
<i>"Smart Solution for Building Crowd Entropy Reduction" (建筑人流熵减智慧解决方案)</i>		
The preceding edition of the same competition series; led the team as first author and won awards in two consecutive editions. The work focused on crowd-density monitoring and smart egress guidance, consistent with the research line on emergency evacuation simulation.		
Fuzhou University Outstanding Undergraduate Thesis	University	Jun 2026
Sole author		
<i>"Pyrolysis and Combustion Experiments and Numerical Simulation of Flame-Retardant Bamboo" (阻燃竹材的热解和燃烧实验及数值模拟研究)</i>		
Combined experiments with numerical simulation to study the pyrolysis characteristics and combustion behaviour of flame-retardant bamboo.		
National Inspirational Scholarship / University Special Scholarship	Multiple awards (Top 1%)	
		2022
Recognising outstanding academic and research performance.		
Fuzhou University "Triple-A" Student	University	

RESEARCH PLAN

Research Plan

1 Research Question

As battery energy storage and renewable-energy infrastructure are deployed at scale, the triggering mechanism, propagation path and control boundary of cell thermal runaway remain poorly defined. The core question is: for a given storage enclosure layout and thermal management regime, how does single-cell thermal runaway propagate at module and enclosure scale, and what minimum safety spacing and mitigation strategy can engineering practice tolerate?

Storage safety Thermal runaway propagation Battery fire

2 Methods and Approach

I will first use Thermakin to characterise the pyrolysis and combustion heat-release behaviour of device-level cell materials, then build module- to enclosure-scale fire field models with the Fire Dynamics Simulator (FDS) to capture the cross-scale propagation of thermal runaway. Model parameters will be calibrated against reduced- and mid-scale combustion experiments, and quantitative risk assessment (QRA) will translate the simulation output into decision-ready risk metrics and safety-spacing criteria. My undergraduate thesis on the pyrolysis and combustion of flame-retardant bamboo already validated this "device-level material characterisation → numerical extrapolation → risk quantification" workflow.

FDS Thermakin QRA Device-level characterisation Experimental calibration

3 Expected Outcomes

A reusable fire risk assessment method for energy storage: criteria for enclosure-scale thermal runaway propagation and defensible safety-spacing recommendations, targeting SCI journal submission, with a method that can serve engineering design and standard-setting rather than remaining a set of numerical case studies; the methods chain can be transferred to the supervisor's ongoing storage-safety or fire-science projects.

Journal articles Safety-spacing criteria Engineering standards

4 Why Australia

Australia has one of the highest renewable-energy penetration rates in the world together with large-scale storage deployment. Grid-scale battery projects in South Australia and Western Australia offer near full-scale data and validation settings for thermal runaway propagation research, and Australian groups have a long record in fire science and storage safety — making it the most suitable environment in which to validate and extend this method.

Australian energy transition

Grid-scale storage

Field validation

5 Three-year Research Timeline

Year 1 (H2 2026–2027): device-level material pyrolysis characterisation and model construction. Use Thermakin to characterise the pyrolysis and heat-release behaviour of target battery/M-rGO materials, build an FDS single-cell thermal-runaway field model, complete a systematic literature review and define the reduced-scale experimental plan. Year 2 (2027–2028): module-scale thermal-runaway propagation simulation and experimental calibration. Extend to module-scale FDS field models, calibrate and validate against the supervisor group's experimental data (subsonic combustion wind tunnel / reduced-scale combustion experiments), and target the first SCI paper submission. Year 3 (2028–2029): enclosure-scale safety-spacing criteria and QRA engineering. Extend to enclosure scale, use QRA to translate simulation results into safety-spacing criteria and engineering-standard recommendations, target the second SCI paper submission, and transfer the method to the supervisor's ongoing projects.

Device-level characterisation

Module-scale propagation

Enclosure-scale criteria

SCI submission planned

TARGET SUPERVISORS

Target Supervisors

Professor Mainak Majumder

ARC Research Hub for Advanced Manufacturing with 2D Materials (AM2D) · Department of Mechanical and Aerospace Engineering, Monash University

Carbon-based energy storage materials and devices: graphene supercapacitors, lithium–sulfur batteries, 2D-material advanced manufacturing. Published M-rGO multiscale curved graphene supercapacitor work in Nature Communications (Sep 2025) — two-step rapid thermal annealing at 700 °C, operando electrochemical interlayer expansion (e-IE) tripling capacitance, 91% retention over 50,000 cycles at 99.7% Coulombic efficiency, 99.5 Wh/L volumetric energy density; published fast-charging Li–S battery work in Advanced Energy Materials (Nov 2024). A "LiS Battery and ultracapacitor hybrid device" project is listed on his homepage as not yet started.

I have read your 2025 Nature Communications paper — M-rGO achieves high volumetric energy density via two-step 700 °C thermal annealing, and the e-IE process lets organic ammonium/ionic-liquid ions enter curved graphene galleries, tripling capacitance with 91% retention over 50,000 cycles. The paper focuses on material-level electrochemical performance; device-level thermal-runaway boundaries and safety-spacing criteria are not yet established — exactly the gap my methods chain can fill: Thermakin to characterise the pyrolysis and heat-release behaviour of M-rGO electrode materials, FDS to build a device-level thermal-runaway propagation field model, and QRA to translate the simulation results into safety-spacing and thermal-boundary criteria. The "LiS Battery and ultracapacitor hybrid device" project on your homepage could host my device-level thermal-safety modelling contribution. I already hold a Monash Master's

Dr Maryam Ghodrat (Senior Lecturer)

Advanced Battery Safety Group (ABS) · School of Engineering and Technology, UNSW Canberra

Li-ion battery and large-scale energy storage system safety, thermal runaway mechanisms and propagation, fire CFD/zone modelling, Wildland–Urban Interface fires; founded the Pyrometric Lab (2021) and built a modular subsonic combustion wind tunnel for controlled-flow thermal-runaway propagation experiments.

I have followed your group's work on Li-ion thermal-runaway propagation and large-scale storage safety, in particular the Pyrometric Lab founded in 2021 with its modular subsonic combustion wind tunnel — which produces controlled combustion flows for thermal-runaway propagation experimental validation. My methods chain is highly complementary to this experimental capability: Thermakin to characterise battery-material pyrolysis heat release as FDS input boundaries, FDS to build enclosure-scale thermal-runaway propagation field models whose parameters can be calibrated and validated directly against subsonic wind-tunnel data, and QRA to convert the results into safety-spacing criteria. The "experimental calibration → numerical extrapolation → risk quantification" chain is isomorphic with the group's "experiment + CFD" research paradigm.

offer in sustainable energy and have paid the deposit, so I am familiar with the faculty's research ecosystem.

OUTREACH & IMPACT

Outreach & Societal Impact

Safety Science Public Communication

Contracted Creator, Guanghe Programme (Education)

2022.09 - Present

Kuaishou platform (Guanghe Programme education-signed creator)

Conducted safety science public communication via short-video platform, producing science content on fire safety, energy-storage safety, and everyday risk prevention, translating safety-engineering research and risk-assessment methods into public-facing safety education; the Guanghe Programme education-signed creator status reflects the platform's recognition of content professionalism.

5M+300K+~30K

Total viewsLikesFollowers

SKILLS

Skills

Numerical Simulation

FDS (fire field modelling)

Pathfinder (evacuation simulation)

Thermakin (fire prediction, Univ. of Maryland)

AutoCAD / 3D modelling

Research Methods & Risk Assessment

SCL (safety check list)

JHA (job hazard analysis)

LEC / LS risk grading

QRA (quantitative risk assessment)

Standards & Management Systems

ISO 9001 Certified Internal Auditor

ISO 14001 / ISO 45001

GB 30871-2022

GB/T 33000-2016

Research Tools & Languages

MATLAB (experimental data processing & modelling)

Origin (data visualisation & curve fitting)

AI LLMs for research (GPT/Claude literature review, Copilot code generation)

English (fluent; writes technical reports independently)

PTE Academic overall 60 (equivalent to IELTS 6.5)

Microsoft Office

Academic writing

Foundation in English literature review and scientific writing

Experiment & modelling

Familiar with fire scenario modelling, evacuation simulation and quantitative risk assessment workflows; proficient in MATLAB and Origin for experimental data processing and modelling, adept at using GPT/Claude for English literature review and academic writing, and GitHub Copilot for Python/MATLAB scripting and FDS model parameter tuning

5 Expert 4 Proficient 3 Competent 2 Familiar 1 Beginner

RESEARCH STATEMENT

Research Statement

Safety engineering is not merely prevention — it is systematic scientific research.

Through study and practice, I have moved from a conventional safety-design mindset towards a research path centred on data-driven and model-based analysis, aiming to optimise and intelligentise safety systems.

My undergraduate training has prepared me for direct entry into doctoral research: converting research results into a patent and software copyrights, leading teams to win consecutive national competition prizes, and completing an Outstanding Undergraduate Thesis have given me experience of the full research cycle from problem definition to dissemination.

I want to work in Australia's energy-transition context, using fire dynamics simulation and quantitative risk assessment to address the safety challenges created by large-scale energy storage.

I hold an offer for Monash's Master of Applied Engineering in Sustainable Energy and have paid the deposit, but having completed my undergraduate research cycle I am applying for direct doctoral entry: I want to invest in multi-year independent research on a specific problem in storage safety rather than coursework. My undergraduate-validated methods chain (experimental calibration plus numerical extrapolation) can plug directly into a prospective supervisor's ongoing storage-safety or fire-science project.

Producing research that holds up in journals and can also be deployed in the field.

- Fire dynamics
- Energy storage safety
- Quantitative risk assessment
- Numerical simulation
- Energy systems

REFEREES

Referees

Fuqiang Yang

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